

Liver transplantation outcomes from controlled circulatory death donors: Static cold storage vs in situ normothermic regional perfusion vs ex situ normothermic machine perfusion

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Supplementary Material

1. Normothermic Regional Perfusion (NRP) technique.

Following circulatory arrest after withdrawal of treatment either in the operating theatre complex or in the intensive care unit, a 5 minute 'no touch' period must elapse before death can be verified. The donor is then transferred into the operating room, rapidly draped and prepared for surgery. NRP is then established in one of two ways. The first involves rapid thoraco-laparotomy and rapid cannulation of the aorta or right common iliac artery, and the inferior vena cava or right common iliac vein, with external clamp occlusion of the descending thoracic aorta. The alternative involves rapid cannulation of the right femoral artery and vein, followed by median sternotomy and clamp occlusion of the descending thoracic aorta. In both cases the cannulae were connected to a circuit for extracorporeal oxygenation and a venting canula was placed in the ascending aorta and left open to air to confirm the absence of cranial perfusion.

The NRP circuit and prime solution have been described in detail previously, comprising compound sodium lactate (Hartmann's solution), heparin, sodium bicarbonate and antimicrobials. Once the venting aortic cannula was in place regional perfusion was established with flows of 2.5 to 3L/min for 2 hours before in situ perfusion with cold UW preservation solution.

The decision to use the liver was based subjectively on liver appearance and biochemical factors. ALT was measured in the perfusate at the start of perfusion and again 60 and 120 minutes later during NRP. Livers were accepted with perfusate ALT of up to 500 iu/L (10x the upper limit of normal) and no significant rise between the first and the second hour value. Similarly, perfusate lactate concentrations were also measured every 30 minutes, with fall in lactate as favorable sign, although lactate would not fall to normal levels due to blood returning from non-perfused areas of the body. All the livers retrieved through NRP were subjected to a period of static cold storage till reperfusion in recipient.

2. Normothermic Machine Perfusion

The device used for normothermic perfusion was either the Liver Assist (XVivo, Groningen, Netherlands) or OrganOx metra (OrganOx, Oxford, UK). There were 16 (24%) NMP livers perfused with Liver Assist device and rest with OrganOx metra. Blood group compatible red cells suspended in either succinylated gelatin (Gelofusine, BBraun) or 4.5% Human Albumin solution was the perfusate with added heparin, amino acids, magnesium, calcium, meropenem and fluconazole. Throughout NMP, livers received continuous infusions of heparin, taurocholic acid, insulin and epoprostenol. Glucose infusion was started when perfusate glucose was <10mmol/L. Perfusate samples were taken at 15, 30, 45, 60, 90, 120 minutes and every 60 minutes thereafter. Bile was sampled when it appeared in adequate volumes for assay. Bile and perfusate gases were analyzed immediately using a Cobas b221 benchtop analyzer (Roche); other biochemistry samples were frozen and stored at -70°C until assayed. The parameters assessed to define suitability of liver function on

NMP are as defined previously by our group. There was no difference in the outcomes of continuous (donor-hospital) NMP and back-to-base NMP (Table S1). However, the comparison was limited by the small numbers of donor-hospital NMP cases (n=5).

Table S1: Comparison of donor-hospital and back-to-base NMP			
Parameter	Donor-hospital (n=5)	Back-to-base (n=62)	P value
Recipient age, years	54 (52 – 59)	59 (51 – 63)	0.424
Model for end stage liver disease (MELD)	11 (10 – 22)	14 (10 – 16)	0.910
Functional Warm ischemia time (WIT), mins	15 (12 – 22)	15 (12 – 18)	0.724
Cold ischemia time (CIT), mins	116 (100 – 132)	400 (360 – 441)	<0.0001
NMP duration, mins	662 (262 – 910)	454 (331 – 566)	0.400
US Donor risk Index (DRI)*	1.9 (1.6 – 2.9)	2.5 (2.0 – 2.9)	0.119
Primary Non-Function (PNF)	0	1 (2)	>0.99
Early Allograft Dysfunction (EAD)#	0	7 (12)	>0.99
Model for Early Allograft Function (MEAF)†	3.3 (1.9 – 4.8)	4.0 (2.6 – 5.8)	0.418
Peak ALT, 1-7 days, U/L	419 (223 – 740)	344 (213 – 740)	0.786
Acute Kidney Injury (AKI)‡	2 (40)	24 (39)	>0.99
Anastomotic biliary stricture	2 (40)	11 (19)	0.273
Non-anastomotic biliary stricture (NAS)	0	7 (12)	>0.99
Graft survival, 1-year	100%	90%	
Patient survival, 1-year	100%	94%	
Values are number (percentage) or median (IQR). Mann-Whitney test for continuous data and Fisher's exact test for categorical data *Feng et al. Am J Transplant, April 2006 #Olthoff et al. Liver Transplantation 2010; 16:943 †Pareja et al. Liver Transplantation 2015; 21:38–46 ‡RIFLE criteria: Peak creatinine ≥ 2 times of baseline in first 7 days or renal replacement therapy (RRT)			

Table S2: Mean ALT during first two hours of normothermic perfusion	
NRP	55 (21 – 131)
NMP	1700 (1093 – 2312)
Values are median (IQR)	
ALT: Alanine transaminase, U/L	